Paul Parisot

Software Engineer - Back-end C++ & Data Systems

Github: paulogarithm — paul.parisot@epitech.eu

About

Rigorous software engineer dedicated to high-performance C++ backend architectures, relational data processing, and Software Craftsmanship.
Strong background in multi-threaded environments, clean code refactoring, TDD methodologies, and Python automation.
Accustomed to demanding industrial systems requiring low-latency computation, deterministic execution, and zero-leak memory safety.

Skills

Core Languages: C, C++ (C++14/17/20), Python, Bash/Shell, Go, Assembly
Databases & Data: SQL, Query Optimization, Database Schema Design, Relational Algebra
Quality & Engineering: TDD (Test-Driven Development), Unit Testing (CPPUnit paradigms), CI/CD, Code Auditing
Architecture: Multithreading, Asynchronous I/O (`Boost.Asio`), Memory Pools, Linux & Windows Development

Professional Experience

Faurecia / Forvia - Software Engineer (Embedded C++ / Agile)

- Developed robust back-end C++ network modules for real-time telemetry, handling high-throughput data streams.
- Implemented strict **TDD (Test-Driven Development)** practices to maximize unit test coverage and prevent regressions.
- Maintained cross-team feature roadmaps in an industrial Agile/Scrum environment using **Jira** tracking.
- Delivered clean, modular architectures praised for their strict handling of complex execution edge-cases.

Gno.land - Core Software Engineer (Distributed Back-end)

- Optimized and refactored core infrastructure modules in a multi-threaded distributed runtime environment.
- Tracked, isolated, and fixed **6 critical runtime crash bugs** and concurrency race conditions, delivering 10k LOC.
- Formulated end-to-end integration test suites integrated into strict automated validation pipelines.

Epitech / IONIS - Systems Programming Instructor & Code Auditor

- Mentored **400+ students** in advanced software engineering concepts, memory layouts, and algorithmic complexity.
- Audited large-scale student repositories to enforce industry-standard quality metrics, error tracking, and clean code principles.

Technical Projects & Data Engineering

Sledge – Cross-Cloud Data Migration Engine (Graduation Project)

Co-engineered a high-reliability data migration platform designed to transfer enterprise assets safely across cloud systems.
Leveraged core relational algebra concepts to implement generic, database-agnostic interface layers.
Wrote and optimized complex structural data queries to ensure strict transactional safety and maximum operational throughput.

Skills: SQL Optimization, Database Schema Design, Relational Algebra, Back-end Infrastructure, System Automation

rodox – Performance-Critical Systems Engine

Built a modular runtime environment under Linux utilizing a strict data-oriented layout (ECS) to prevent CPU cache misses.
Engineered custom **static pool allocators** to entirely bypass runtime heap allocation latency (`malloc`/`new`).
Integrated `Boost.Asio` for high-efficiency asynchronous execution flows alongside hardware-accelerated vectorization.

Skills: C++, Boost.Asio, Custom Memory Management, Cache Locality, Multithreading, Asynchronous Frameworks

Advanced Theoretical & Systems Core

- **Computer Science Foundations**: Deep preparation for high-level technical competitions, specializing in advanced concurrency control theories, data normalization, and language operational semantics.
- **Runtime Memory Mutation (cclass)**: Built a pure C runtime framework that patches compiled assembly stubs on-the-fly inside execution-allocated `mmap` pages, achieving clean object-oriented syntax without virtual table overhead.

Education

Epitech, Paris 2022 – 2027
Master's Degree in Information Technology (Master's level diploma)
Level-7 Certification - Expert in Software Engineering
Specialization: Systems Architecture

McGill University, Montréal 2025 – 2026
Certificate in Management
Technical Leadership & Strategy